

# Rahul Rawat

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## PROFILE

Data Science Master's student at SRH Heidelberg with hands on experience designing data pipelines for sensor and time series data and building machine learning models for forecasting and anomaly style problems. Has built a real time pipeline polling live data every 30 seconds and joining it against multiple enrichment sources, and has benchmarked classical and gradient boosted models including XGBoost and CatBoost for production style forecasting tasks. Comfortable with Python, SQL, and cloud platforms, and confident explaining model behavior to non technical stakeholders using tools like SHAP. Excited to bring that data engineering and data science foundation to Mercedes-Benz Tech Innovation's AI for vehicle validation team.

## EDUCATION

### M.Sc. Data Science and Analytics

Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

### Bachelor's Degree in Computer Science

2019 to 2023

GL Bajaj Institute of Technology and Management

## EXPERIENCE

### eRay GmbH

Oct 2025 to Mar 2026

Data Scientist

- Built a complete recursive forecasting pipeline covering four water quality indicators for a German lake: chlorophyll a, turbidity, pH, and dissolved oxygen, across a six month academic and industry collaboration.
- Benchmarked six models head to head, including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, then used CatBoost multi quantile regression to deliver asymmetric 80 percent prediction intervals.
- Enforced strict rules against data leakage, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not predictable without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast curve to stop tree models flatlining during recursive prediction.
- Wrapped the pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade.

## PERSONAL PROJECTS

**Real Time Flight Tracking Data Pipeline:** Python, PySpark, BigQuery, dbt, Apache Airflow, Google Cloud Platform, Tableau

- Set up real time data collection from the OpenSky Network API, polling live position data every 30 seconds and enriching each reading against four separate reference and sensor style sources.
- Cleaned raw data with PySpark on Google Cloud and shaped it into analysis ready tables with dbt, computing derived features such as each aircraft's nearest airport along the way.
- Orchestrated the whole system with Apache Airflow so the batch and real time layers refresh automatically every 15 minutes, the same pattern used for continuously monitoring streaming sensor data.
- Surfaced findings in Tableau using Python statistics through TabPy, showing that traffic volume dropped 4.4 times over in heavy rain, an anomaly style pattern detected directly from the data.

**CreditIQ, a Fairness by Design Credit Scoring System:** Python, Scikit Learn, AIF360, SHAP, Streamlit

- Lifted the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88, clearing the EU AI Act and AGG 80 percent fairness threshold.
- Diagnosed a hidden intersectional bias, where younger women were penalized even after age bias was corrected, using SHAP, then designed a four way age by gender threshold matrix to fix it without over correcting into reverse discrimination.
- Cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, documenting the fairness versus accuracy trade off as a deliberate, defensible decision rather than an accident.
- Backed the pipeline with unit tests at 100 percent branch coverage and a full written regulatory review covering the EU AI Act, GDPR, a model card, and attack vectors.

**Movie Analytics and ML Pipeline, a Cloud Native Data Platform:** Google Cloud Platform, BigQuery, Cloud Run, Cloud Scheduler, Python, SQL, Looker Studio

- Built a complete batch pipeline that pulls movie data from a public API into a GCS data lake, processes it through Bronze to Silver to Gold medallion architecture in BigQuery, and runs fully automated on a Cloud Scheduler trigger with no manual steps.
- Engineered the Silver layer for trustworthy analytics, covering schema enforcement, safe type casting, deduplication through window functions, and genre normalization into a relational model.
- Trained a BigQuery ML classifier to predict whether a film will be a hit before release, deliberately splitting features into two tables to prevent data leakage by using only pre release signals.
- Secured the system with a least privilege service account and Secret Manager, and version controlled all code, SQL, and schemas in a reproducible GitHub repo.

## RESEARCH AND THESIS

**Bachelor Thesis: Diabetes Prediction Using Machine Learning**

*Python, Scikit Learn, Pandas, Seaborn, Google Colab. Written up as an IEEE style paper.*

- Built a complete machine learning pipeline comparing six classifiers on a clinical dataset of 768 patient records, using 10 fold cross validation and per model confusion matrices.
- Caught biologically impossible zero values the original authors had overlooked, then applied IQR based outlier removal and proper imputation.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure.
- Wrote up findings as an IEEE style paper, including an honest section on limitations and what to do differently.

## LANGUAGES

**English:** fluent, professional working proficiency

**German:** A2.2, completed through VHS